

# AMAR DHILLON

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## Education

### University of Maryland, College Park

*B.S. in Computer Science, Minor in Statistics; GPA: 3.5*

Expected May 2028

*College Park, MD*

## Experience

### Amazon Web Services

Sep 2026 – Dec 2026

*Software Development Engineer Intern*

*Redmond, WA*

- Writing backtested Java capacity-planning logic that sets upper and lower bounds on the EC2 capacity Amazon Redshift reserves for patching to balance availability against cost

### Capital One

Jun 2026 – Aug 2026

*Software Engineering Intern*

*Richmond, VA*

- Rebuilt a legacy test-data workflow tool as a React and TypeScript Electron app with a visual React Flow editor and Jest-tested validation to make editing smoother, which **90%** of 30+ surveyed developers preferred
- Cut AI workflow generation time **65%** to under 15s by splitting a Claude skill into focused sub-skills that call Python search and validation tools instead of relying on open-ended model search
- Precomputed popular workflows from a 270K+ row PostgreSQL table into a Redis-cached category tree to keep lookups within the database's permission and rate limits, which became the Claude skill's main search tool

### UMD App Development Contracting (Client: Amazon Leo)

Jan 2026 – Present

*Software Engineer (Contract)*

*College Park, MD*

- Co-built a topic-scoring research tool and deployed it to dev on AWS Fargate with GitHub Actions CI/CD and AWS CDK to show Amazon Leo's public-policy staff which research topics are popular
- Reduced end-to-end latency **34%** for a pipeline routing 6,000+ scraped research documents to inference by replacing S3 staging hops with Kinesis streams feeding Lambda
- Built an EventBridge-scheduled LLM-as-judge check for a production Amazon Bedrock classifier that rates research documents by importance to trigger fine-tuning only when its grades shift past a threshold

### UMD Institute for Advanced Computer Studies (UMIACS)

Dec 2025 – Present

*Machine Learning Research Intern*

*College Park, MD*

- Built and prompt-tuned a pipeline that overlays AnyGrasp's candidate grasps on camera images in Open3D to help a vision-language model pick robot grasps that avoid hitting the table for a PhD-led project targeting ICLR

### TerpLabs

Jun 2025 – Present

*Co-Founder, Software Engineer*

*College Park, MD*

- Co-founded a 110+ member student product studio that has shipped 5 live products
- Built Redis caching in front of Tortuga's Prisma ORM queries to reduce database load and latency for a class scheduler that drew 20,000+ visits in its first 30 days
- Building RAG meal suggestions for the official UMD app with SentenceBERT vector search in ChromaDB to feed GPT-4o mini real scraped recipes, which internal testers rated satisfactory for **93%** of meals

## Projects

### IEX Order Book & Matching Engine | C++17, STL, POSIX, Linux perf

Jul 2026 – Aug 2026

- Built a C++17 order book and matching engine to replay **50.9M** real IEX exchange messages without ever resyncing, which agreed with the exchange on 99,122 of 99,123 fills and 100% of top-of-book prices
- Implemented B+ tree and LSM-tree backends for the engine's event log to compare them on identical 2M-record workloads, which showed the B+ tree giving **12.8x** the read throughput and 22x faster range queries for 27% more disk
- Ran the full replay at **1.26M events/sec** end-to-end on an order book that keeps std::list FIFOs per price level in a std::map with an order-ID hash map to make cancels O(1)

## Technical Skills

**Languages:** Python, Java, C++, C, TypeScript, JavaScript, SQL, Bash

**Cloud & DevOps:** AWS (Lambda, Kinesis, Fargate, EC2, API Gateway, EventBridge, CloudWatch), AWS CDK (infrastructure as code), Docker, GitHub Actions (CI/CD)

**Backend & Full-Stack:** REST APIs, Node.js, React, Electron, PostgreSQL, Redis, SQLAlchemy, Prisma, OpenSearch, ChromaDB

**AI/ML:** LLMs (Claude, Bedrock, GPT-4o mini), retrieval-augmented generation (RAG), fine-tuning and evaluation, PyTorch

**Practices & Tools:** data structures and algorithms, unit testing (Pytest, Jest), profiling (perf, psutil), Linux, Git, Jira, Agile

**Certification:** AWS Certified Developer – Associate (Nov 2024 – Nov 2027)